Results of the Original Model Under the Proposed Methodology

Eduardo Henrique Basilio de Carvalho

June 3, 2026

ARI Mean (All)

Dataset	CluStream	DenStream	MicroTEDAClus
Benchmark(1)	0.326 ± 0.087	0.403 ± 0.278	$\mathbf{0.437 \pm 0.163}$
Benchmark(2)	$\mathbf{0.582 \pm 0.022}$	0.347 ± 0.035	0.491 ± 0.249
RandomRBFGeneratorEvents	0.199 ± 0.114	$\mathbf{0.374 \pm 0.156}$	0.359 ± 0.194
cassini	$\mathbf{0.544 \pm 0.040}$	0.281 ± 0.035	0.420 ± 0.213
chameleon_ds7	$\mathbf{0.325 \pm 0.048}$	0.159	0.232 ± 0.033
chameleon_ds8	$\mathbf{0.329 \pm 0.042}$	0.165 ± 0.044	0.205 ± 0.033
SCL-1	0.625 ± 0.154	0.063 ± 0.077	$\mathbf{0.644 \pm 0.166}$
SCL-2	0.743 ± 0.132	0.430 ± 0.195	$\mathbf{0.832 \pm 0.010}$
SCL-4	0.739 ± 0.175	0.279 ± 0.220	$\mathbf{0.826 \pm 0.040}$
SCL-8	0.861 ± 0.088	0.250 ± 0.191	$\mathbf{0.870 \pm 0.034}$
SCL-16	$\mathbf{0.871 \pm 0.065}$	0.155 ± 0.214	0.687 ± 0.341
SCL-32	$\mathbf{0.921 \pm 0.034}$	0.000	0.681 ± 0.184
SCL-64	$\mathbf{0.894 \pm 0.068}$	0.000	0.509 ± 0.285
SCL-128	$\mathbf{0.857 \pm 0.104}$	0.000	0.406 ± 0.279

Table: ARI mean (all)

ARI Mean (First 10)

Dataset	CluStream	DenStream	MicroTEDAClus
Benchmark(1)	0.456 ± 0.101	0.000	$\mathbf{0.746 \pm 0.043}$
Benchmark(2)	$\mathbf{0.531 \pm 0.042}$	0.000	0.456 ± 0.234
RandomRBFGeneratorEvents	0.128 ± 0.093	0.000	$\mathbf{0.520 \pm 0.189}$
cassini	$\mathbf{0.493 \pm 0.077}$	0.000	0.376 ± 0.240
chameleon_ds7	$\mathbf{0.263 \pm 0.040}$	0.000	0.090 ± 0.036
chameleon_ds8	$\mathbf{0.264 \pm 0.030}$	0.000	0.119 ± 0.036
SCL-1	0.536 ± 0.135	0.000	$\mathbf{0.570 \pm 0.173}$
SCL-2	0.680 ± 0.131	0.000	$\mathbf{0.773 \pm 0.021}$
SCL-4	0.652 ± 0.164	0.000	$\mathbf{0.781 \pm 0.016}$
SCL-8	0.787 ± 0.079	0.000	$\mathbf{0.809 \pm 0.032}$
SCL-16	$\mathbf{0.780 \pm 0.059}$	0.000	0.731 ± 0.169
SCL-32	$\mathbf{0.840 \pm 0.040}$	0.000	0.824 ± 0.051
SCL-64	$\mathbf{0.820 \pm 0.057}$	0.000	0.678 ± 0.327
SCL-128	0.668 ± 0.259	0.000	$\mathbf{0.696 \pm 0.288}$

Table: ARI mean (first 10)

ARI Mean (Last 10)

Dataset	CluStream	DenStream	MicroTEDAClus
Benchmark(1)	0.211 ± 0.153	$\mathbf{0.626 \pm 0.432}$	0.176 ± 0.344
Benchmark(2)	$\mathbf{0.591 \pm 0.102}$	0.579 ± 0.032	0.460 ± 0.251
RandomRBFGeneratorEvents	0.267 ± 0.119	$\mathbf{0.415 \pm 0.243}$	0.177 ± 0.258
cassini	$\mathbf{0.570 \pm 0.044}$	0.418 ± 0.067	0.425 ± 0.193
chameleon_ds7	$\mathbf{0.326 \pm 0.065}$	0.182	0.187 ± 0.044
chameleon_ds8	$\mathbf{0.353 \pm 0.054}$	0.255 ± 0.041	0.163 ± 0.054
SCL-1	0.662 ± 0.161	0.055 ± 0.089	$\mathbf{0.662 \pm 0.168}$
SCL-2	0.762 ± 0.151	0.624 ± 0.286	$\mathbf{0.855 \pm 0.016}$
SCL-4	0.772 ± 0.181	0.409 ± 0.320	$\mathbf{0.818 \pm 0.130}$
SCL-8	0.881 ± 0.129	0.379 ± 0.303	$\mathbf{0.892 \pm 0.049}$
SCL-16	$\mathbf{0.903 \pm 0.094}$	0.228 ± 0.315	0.644 ± 0.450
SCL-32	$\mathbf{0.951 \pm 0.052}$	0.000	0.431 ± 0.400
SCL-64	$\mathbf{0.928 \pm 0.083}$	0.000	0.162 ± 0.286
SCL-128	$\mathbf{0.942 \pm 0.090}$	0.000	0.185 ± 0.363

Table: ARI mean (last 10)

ARI Diff (Max-Last)

Dataset	CluStream	DenStream	MicroTEDAClus
Benchmark(1)	0.535 ± 0.416	$\mathbf{0.078 \pm 0.087}$	0.761 ± 0.337
Benchmark(2)	$\mathbf{0.268 \pm 0.190}$	0.382 ± 0.099	0.392 ± 0.203
RandomRBFGeneratorEvents	0.437 ± 0.359	$\mathbf{0.399 \pm 0.291}$	0.622 ± 0.244
cassini	$\mathbf{0.133 \pm 0.104}$	0.154 ± 0.040	0.226 ± 0.139
chameleon_ds7	0.218 ± 0.076	$\mathbf{0.184}$	0.312 ± 0.068
chameleon_ds8	0.116 ± 0.046	$\mathbf{0.008 \pm 0.026}$	0.311 ± 0.034
SCL-1	0.140 ± 0.077	0.240 ± 0.306	$\mathbf{0.136 \pm 0.075}$
SCL-2	0.177 ± 0.159	$\mathbf{0.077 \pm 0.075}$	0.095 ± 0.054
SCL-4	0.117 ± 0.106	$\mathbf{0.053 \pm 0.046}$	0.112 ± 0.134
SCL-8	0.104 ± 0.133	$\mathbf{0.074 \pm 0.066}$	0.088 ± 0.065
SCL-16	0.140 ± 0.194	$\mathbf{0.042 \pm 0.070}$	0.442 ± 0.437
SCL-32	0.049 ± 0.039	$\mathbf{0.000}$	0.621 ± 0.483
SCL-64	0.079 ± 0.151	$\mathbf{0.000}$	0.804 ± 0.308
SCL-128	0.081 ± 0.168	$\mathbf{0.000}$	0.795 ± 0.381

Table: ARI diff (max-last)

Fit Time (ms)

Dataset	CluStream	DenStream	MicroTEDAClus
Benchmark(1)	8.916 ± 12.930	**4.357 ± 22.400**	43.430 ± 45.460
Benchmark(2)	18.230 ± 29.420	**5.952 ± 30.250**	16.040 ± 11.350
RandomRBFGeneratorEvents	38.110 ± 162.900	**2.285 ± 16.190**	272.600 ± 246.400
cassini	17.050 ± 42.200	**5.805 ± 29.750**	29.650 ± 36.200
chameleon_ds7	5.137 ± 6.847	**2.517 ± 19.100**	238.700 ± 59.230
chameleon_ds8	19.170 ± 26.980	**2.992 ± 20.650**	430.800 ± 648.300
SCL-1	12.340 ± 17.800	**4.745 ± 23.160**	39.470 ± 36.070
SCL-2	8.286 ± 15.600	**5.289 ± 28.850**	97.680 ± 132.500
SCL-4	16.630 ± 25.320	**6.891 ± 38.170**	40.070 ± 44.250
SCL-8	23.670 ± 50.890	**10.570 ± 58.860**	33.610 ± 37.550
SCL-16	82.510 ± 164.100	**17.300 ± 96.840**	20.630 ± 19.270
SCL-32	130.900 ± 140.000	31.490 ± 177.500	**18.950 ± 25.610**
SCL-64	333.200 ± 685.700	61.600 ± 348.000	**13.950 ± 24.690**
SCL-128	6376.000 ± 27 750.000	123.700 ± 699.500	**16.960 ± 29.410**

Table: Fit time (ms)

Predict Time (ms)

Dataset	CluStream	DenStream	MicroTEDAClus
Benchmark(1)	**2.794 ± 1.361**	21.270 ± 50.720	7.762 ± 11.840
Benchmark(2)	2.159 ± 0.912	4.858 ± 6.767	**1.423 ± 0.877**
RandomRBFGeneratorEvents	9.674 ± 7.838	**3.160 ± 3.043**	57.920 ± 80.930
cassini	3.002 ± 1.838	4.481 ± 5.030	**2.747 ± 3.054**
chameleon_ds7	3.026 ± 1.277	**0.817 ± 0.283**	4.552 ± 4.460
chameleon_ds8	3.555 ± 1.618	**1.803 ± 1.040**	28.080 ± 102.900
SCL-1	2.225 ± 0.952	**0.619 ± 1.156**	3.357 ± 3.203
SCL-2	2.296 ± 1.109	**1.843 ± 3.629**	7.276 ± 11.080
SCL-4	3.570 ± 2.322	**0.723 ± 0.489**	3.695 ± 4.292
SCL-8	5.950 ± 3.875	**1.190 ± 2.218**	4.027 ± 4.566
SCL-16	7.283 ± 4.703	**0.890 ± 0.505**	2.449 ± 1.952
SCL-32	13.380 ± 5.561	**1.023 ± 0.319**	1.921 ± 1.517
SCL-64	23.900 ± 16.370	1.705 ± 0.456	**1.423 ± 0.924**
SCL-128	117.300 ± 146.500	3.077 ± 0.732	**1.483 ± 1.240**

Table: Predict time (ms)

Memory (MB)

Dataset	CluStream	DenStream	MicroTEDAClus
Benchmark(1)	0.201 ± 0.111	$\mathbf{0.022 \pm 0.023}$	0.241 ± 0.250
Benchmark(2)	0.145 ± 0.073	$\mathbf{0.014 \pm 0.009}$	0.097 ± 0.068
RandomRBFGeneratorEvents	0.710 ± 0.560	$\mathbf{0.010 \pm 0.005}$	1.021 ± 0.806
cassini	0.214 ± 0.145	$\mathbf{0.014 \pm 0.008}$	0.141 ± 0.119
chameleon_ds7	0.214 ± 0.095	$\mathbf{0.005 \pm 0.001}$	0.488 ± 0.075
chameleon_ds8	0.263 ± 0.130	$\mathbf{0.008 \pm 0.002}$	0.681 ± 0.724
SCL-1	0.146 ± 0.071	$\mathbf{0.009 \pm 0.007}$	0.160 ± 0.119
SCL-2	0.154 ± 0.092	$\mathbf{0.008 \pm 0.006}$	0.373 ± 0.398
SCL-4	0.267 ± 0.192	$\mathbf{0.006 \pm 0.002}$	0.218 ± 0.165
SCL-8	0.515 ± 0.359	$\mathbf{0.008 \pm 0.007}$	0.259 ± 0.193
SCL-16	0.652 ± 0.515	$\mathbf{0.008 \pm 0.004}$	0.168 ± 0.229
SCL-32	1.288 ± 0.634	$\mathbf{0.009 \pm 0.002}$	0.441 ± 0.574
SCL-64	2.275 ± 1.962	$\mathbf{0.014 \pm 0.006}$	0.323 ± 0.818
SCL-128	12.741 ± 15.745	$\mathbf{0.025 \pm 0.013}$	0.917 ± 1.888

Table: Memory (MB)

Memory (Max, MB)

Dataset	CluStream	DenStream	MicroTEDAClus
Benchmark(1)	0.204 ± 0.112	$\mathbf{0.040 \pm 0.027}$	0.347 ± 0.300
Benchmark(2)	0.145 ± 0.074	$\mathbf{0.026 \pm 0.009}$	0.127 ± 0.079
RandomRBFGeneratorEvents	0.734 ± 0.574	$\mathbf{0.013 \pm 0.006}$	1.539 ± 1.073
cassini	0.216 ± 0.148	$\mathbf{0.023 \pm 0.008}$	0.182 ± 0.133
chameleon_ds7	0.215 ± 0.096	$\mathbf{0.009}$	0.528 ± 0.077
chameleon_ds8	0.265 ± 0.130	$\mathbf{0.010 \pm 0.002}$	0.889 ± 1.102
SCL-1	0.147 ± 0.071	$\mathbf{0.026 \pm 0.025}$	0.220 ± 0.158
SCL-2	0.155 ± 0.093	$\mathbf{0.013 \pm 0.008}$	0.549 ± 0.515
SCL-4	0.270 ± 0.195	$\mathbf{0.009}$	0.357 ± 0.208
SCL-8	0.521 ± 0.366	$\mathbf{0.012 \pm 0.009}$	0.425 ± 0.156
SCL-16	0.657 ± 0.520	$\mathbf{0.011 \pm 0.003}$	0.514 ± 0.236
SCL-32	1.293 ± 0.638	$\mathbf{0.010}$	1.065 ± 0.367
SCL-64	2.292 ± 1.981	$\mathbf{0.018}$	1.156 ± 1.184
SCL-128	13.609 ± 17.118	$\mathbf{0.033}$	0.973 ± 1.961

Table: Memory (max) (MB)

Microcluster Count

Dataset	CluStream	DenStream	MicroTEDAClus
Benchmark(1)	100 ± 57	**12 ± 17**	92 ± 104
Benchmark(2)	71 ± 38	**7 ± 7**	31 ± 20
RandomRBFGeneratorEvents	360 ± 286	**5 ± 3**	350 ± 278
cassini	107 ± 75	**7 ± 6**	49 ± 43
chameleon_ds7	107 ± 49	**2 ± 1**	328 ± 58
chameleon_ds8	131 ± 66	**3 ± 2**	554 ± 756
SCL-1	91 ± 46	**4 ± 7**	61 ± 45
SCL-2	76 ± 47	**3 ± 4**	106 ± 123
SCL-4	95 ± 70	**1 ± 1**	56 ± 51
SCL-8	109 ± 77	**2 ± 2**	47 ± 41
SCL-16	76 ± 61	**1 ± 1**	30 ± 24
SCL-32	80 ± 40	**1**	36 ± 58
SCL-64	73 ± 63	**1**	27 ± 59
SCL-128	206 ± 255	**1**	26 ± 57

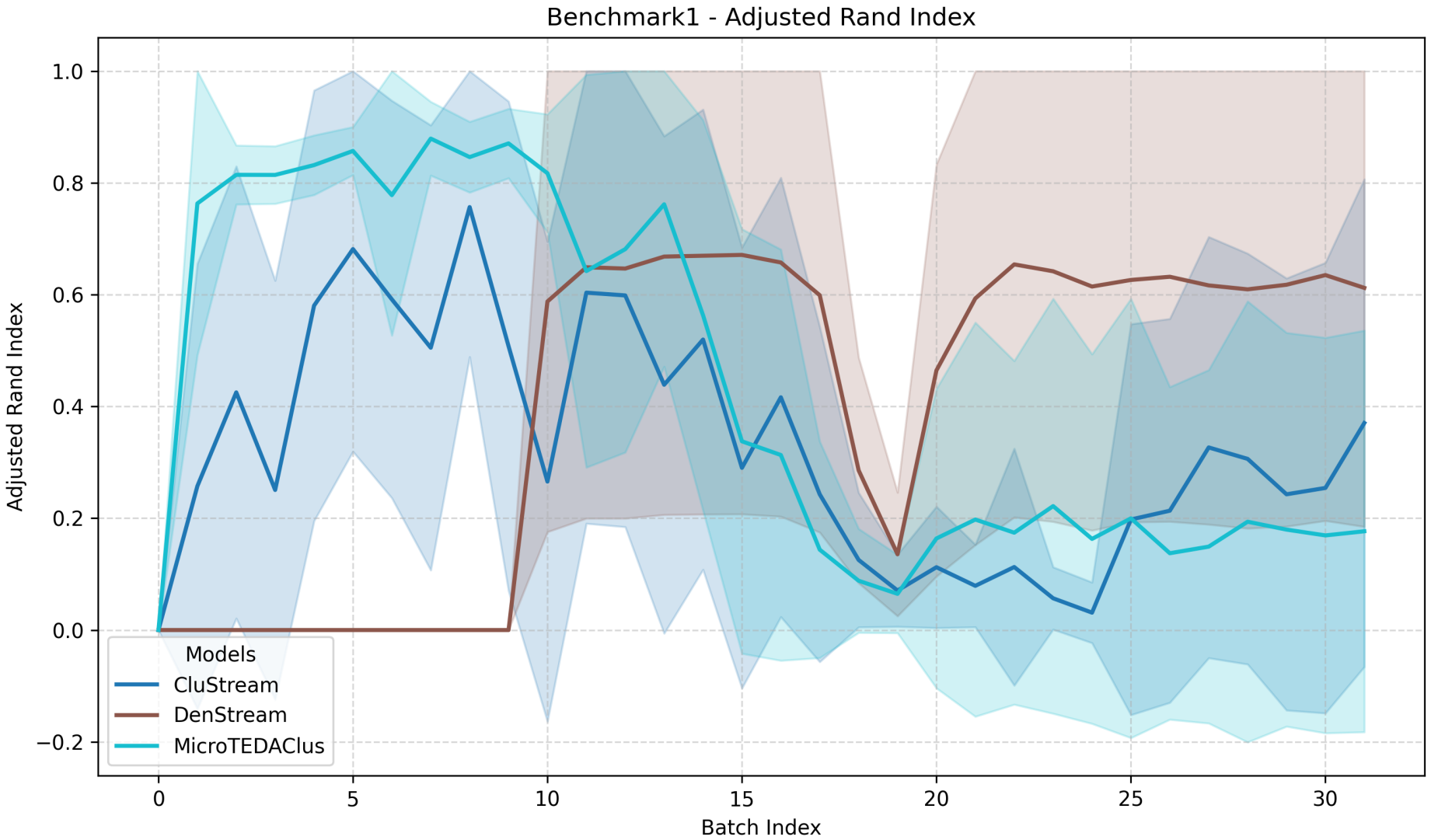
Table: Microcluster count

Microcluster Count (Max)

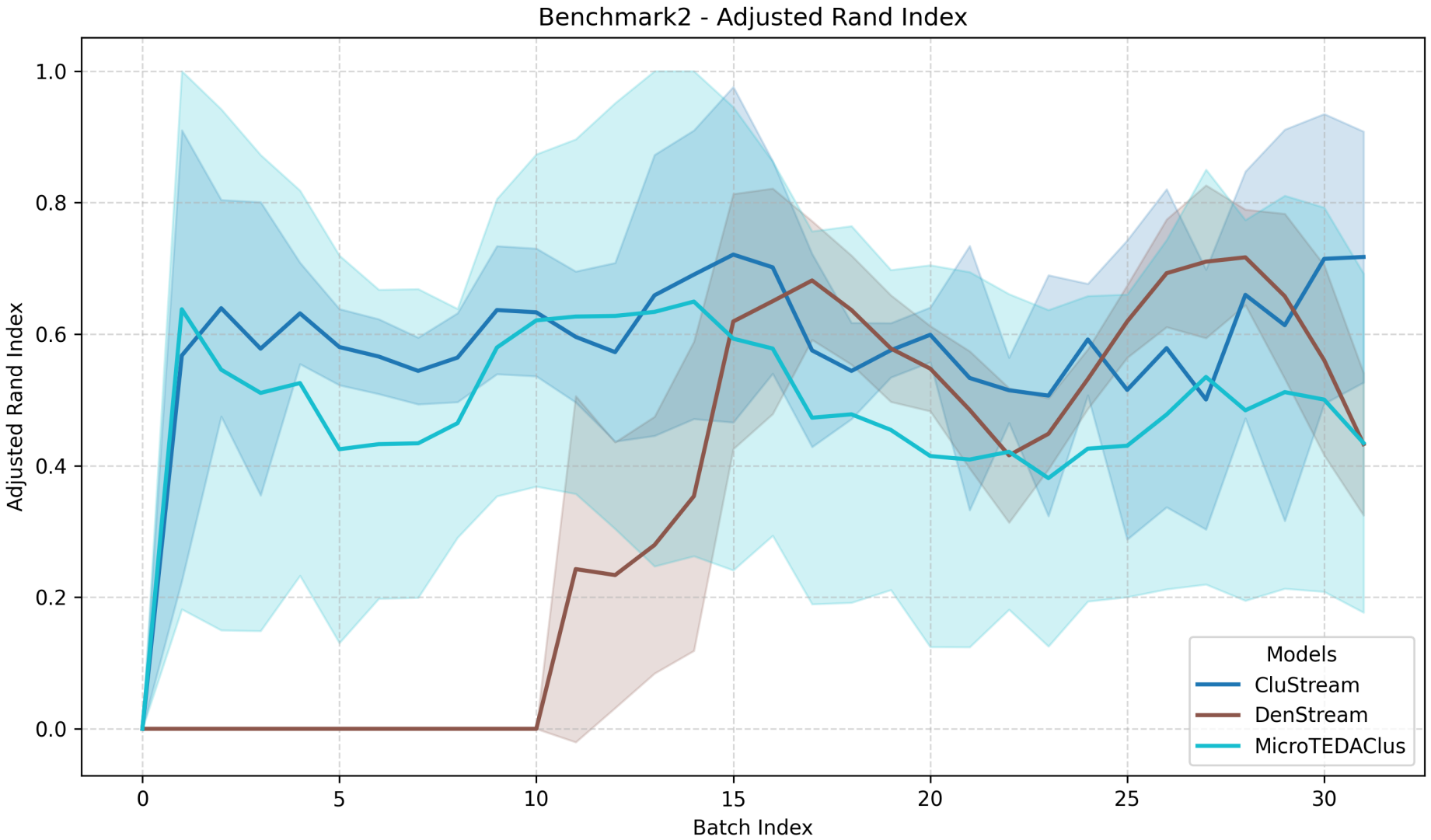
Dataset	CluStream	DenStream	MicroTEDAClus
Benchmark(1)	101 ± 57	**25 ± 20**	132 ± 127
Benchmark(2)	71 ± 38	**16 ± 6**	40 ± 23
RandomRBFGeneratorEvents	371 ± 292	**6 ± 4**	498 ± 338
cassini	108 ± 76	**14 ± 5**	58 ± 45
chameleon_ds7	108 ± 49	**2 ± 1**	356 ± 56
chameleon_ds8	132 ± 67	**4 ± 1**	748 ± 1139
SCL-1	91 ± 46	**20 ± 25**	78 ± 56
SCL-2	76 ± 48	**6 ± 6**	154 ± 163
SCL-4	95 ± 71	**2 ± 1**	86 ± 62
SCL-8	111 ± 79	**3 ± 3**	68 ± 54
SCL-16	77 ± 62	**2 ± 1**	48 ± 27
SCL-32	80 ± 40	**1**	70 ± 95
SCL-64	73 ± 64	**1**	52 ± 95
SCL-128	219 ± 276	**1**	55 ± 94

Table: Microcluster count (max)

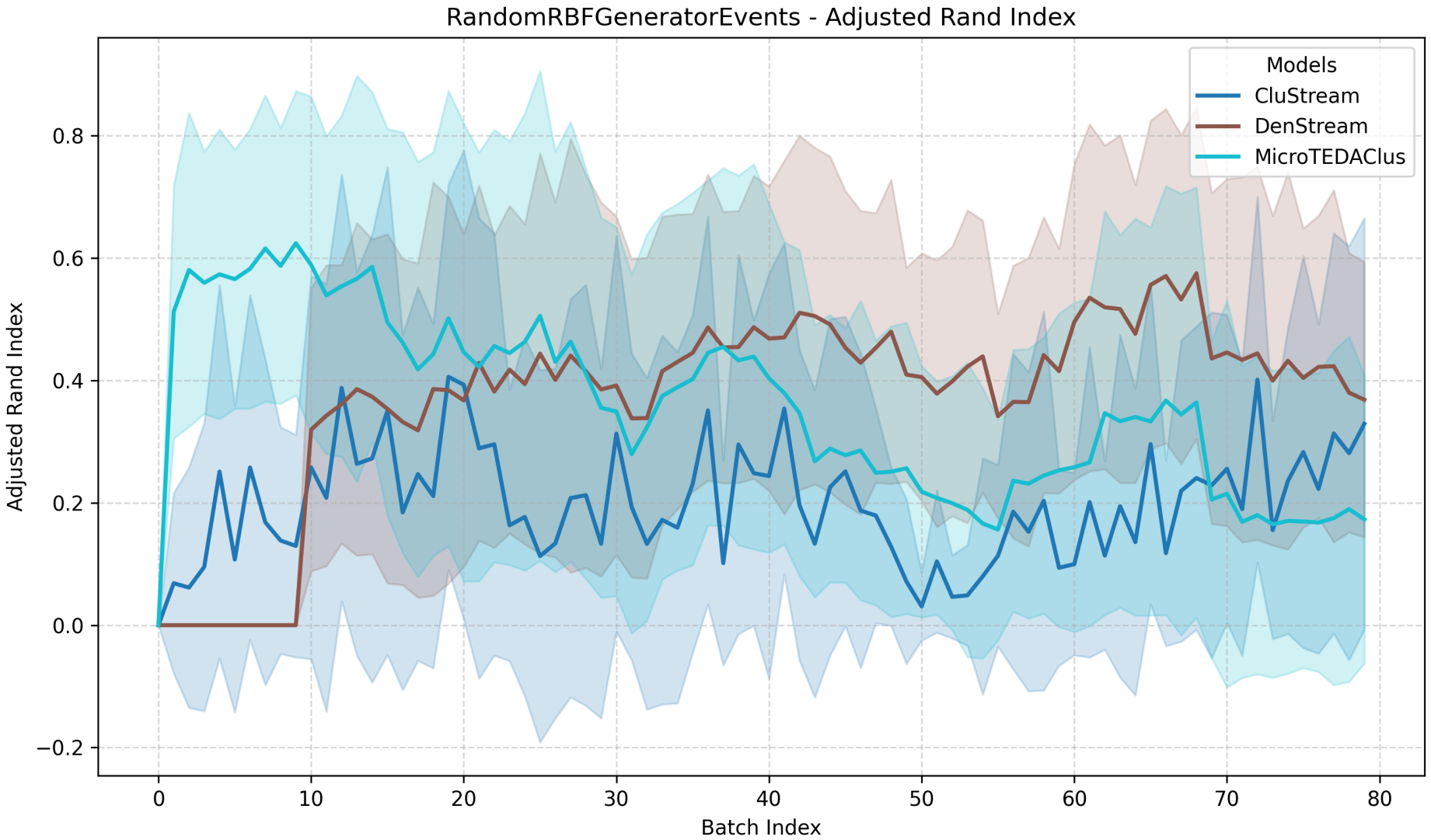
ARI - Benchmark 1



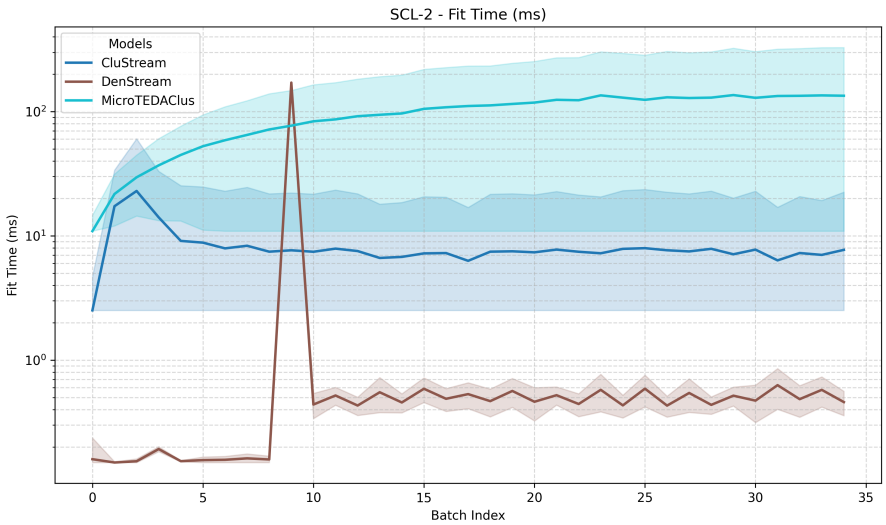
ARI - Benchmark 2

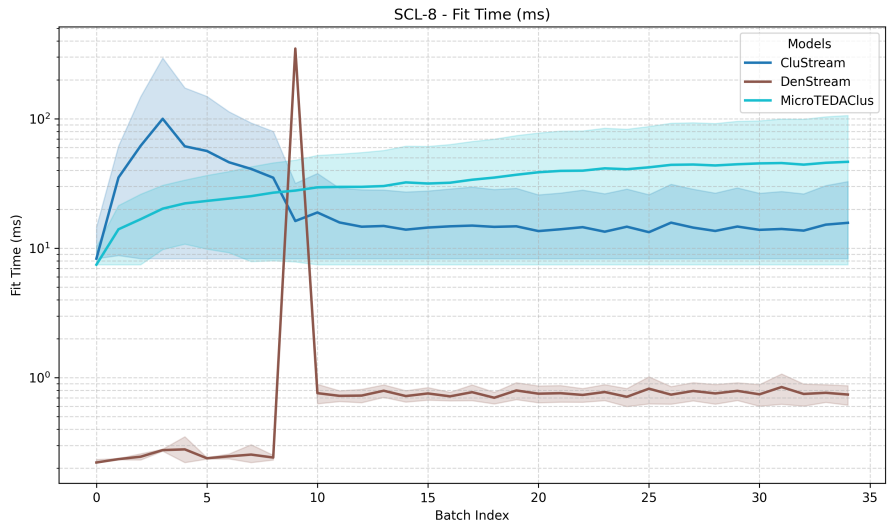


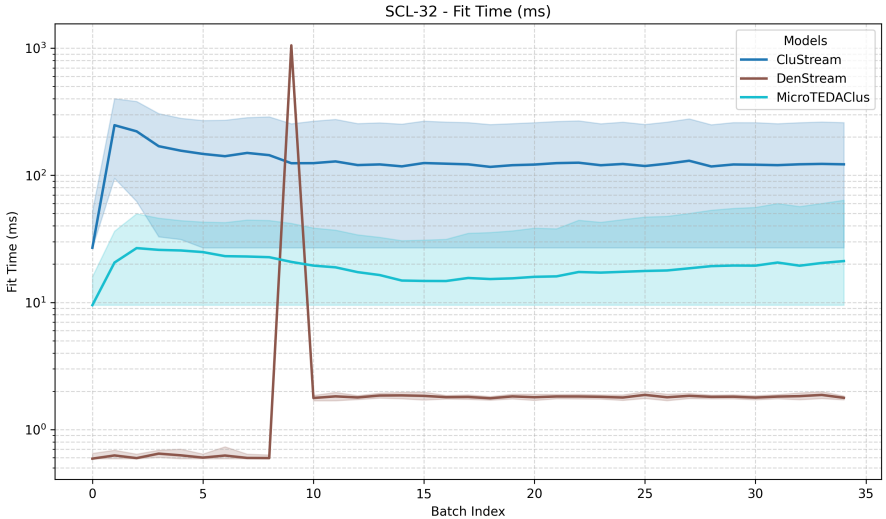
ARI - RandomRBFGeneratorEvents

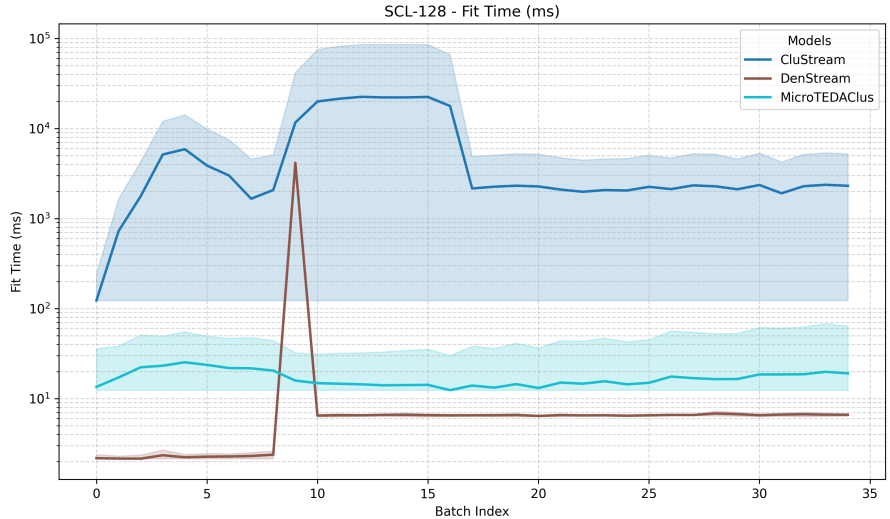


Fit Time - Dimensions

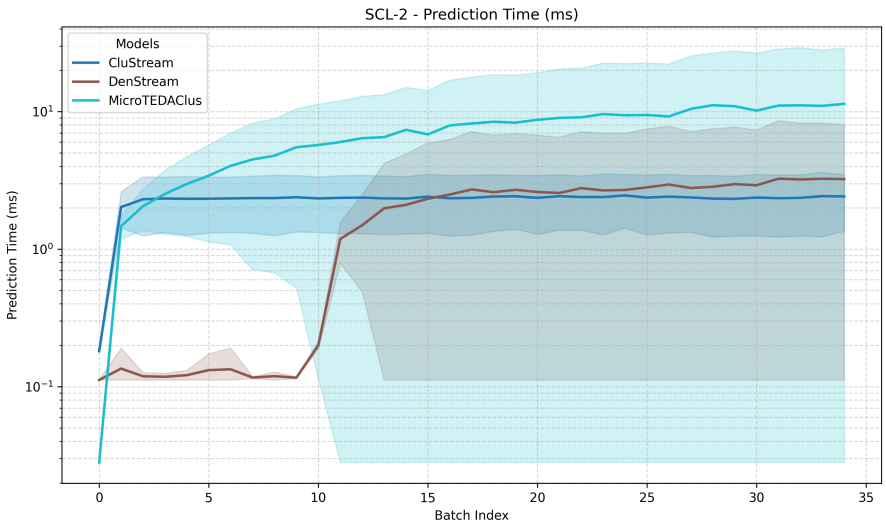


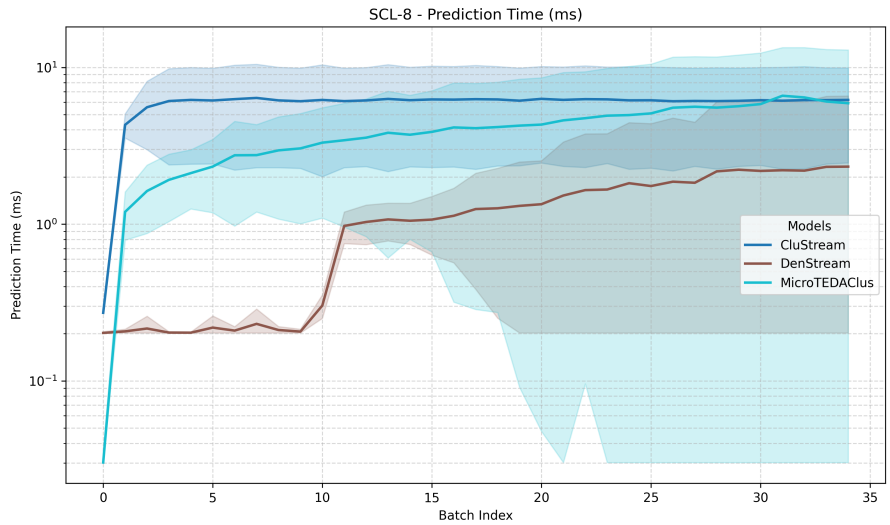


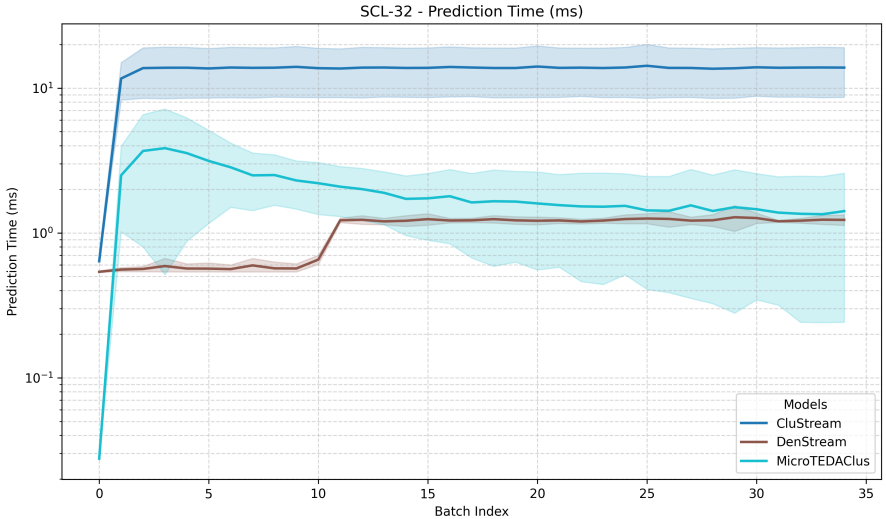


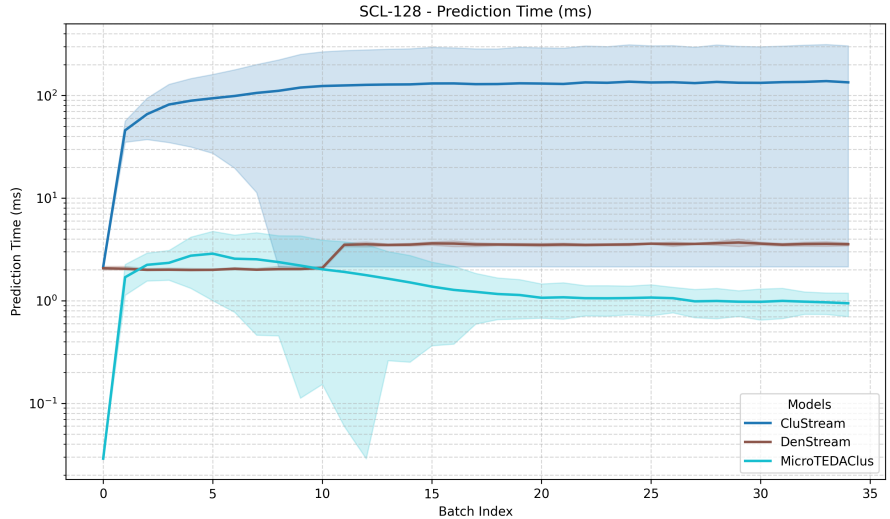


Predict Time - Dimensions

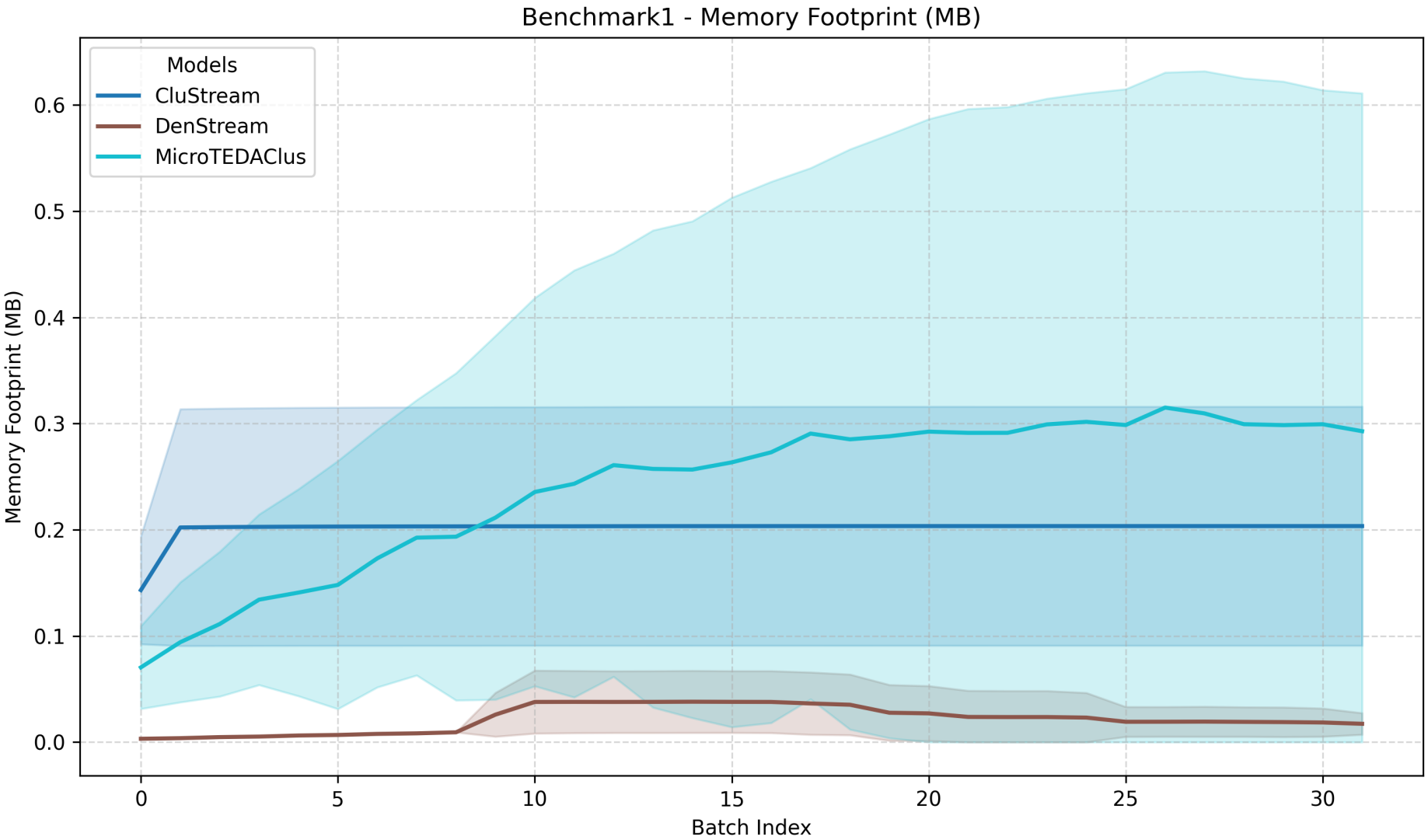








Memory - Benchmark 1



Microcluster Count - Benchmark 1

